

Fuzzy RDD: Airbnb Superhost Effect

Setup

```
library(duckdb)
library(rdrobust)
library(rddensity)
library(ggplot2)
library(gpss)
library(tidyr)
library(stringr)
```

Load Data

```
con      <- dbConnect(duckdb(), "../data/listings.db")
listings_rdd <- dbGetQuery(con, "SELECT * FROM listings_fuzzy_rdd")
dbDisconnect(con)
```

Center Running Variable

```
# Center around the 4.8 superhost cutoff
listings_rdd$centered_overall_score <- listings_rdd$avg_host_review_score - 4.8

# last_scraped is aliased as scrape_date in the dbt pipeline
listings_rdd$host_tenure_days <- as.numeric(
  as.Date(listings_rdd$scrape_date) - as.Date(listings_rdd$host_since)
)
```

We center so that the regression model is simpler to interpret. The causal effects will still be the same!

Uncentered model:

$$\text{reviews last 12 months} = \alpha + \beta_1 \text{has superhost} + \beta_2 \text{overall ratings} + \epsilon$$

With an uncentered model, to calculate reviews in the last 12 months for someone at the cutoff who isn't a superhost, we would have to do $\alpha + 4.8\beta_2 + \epsilon$.

Centered model:

$$\text{reviews last 12 months} = \alpha + \beta_1 \text{has superhost} + \beta_2 \text{centered overall ratings} + \epsilon$$

With the centered model, to calculate the same value, we would only have to do $\alpha + \epsilon$ since the β_2 will become 0, since the cutoff of the centered ratings is 0.

Exploratory RDD Plot

Before running a formal analysis, we want to see if there is even a discontinuity.

```
cutoff <- 0
nbins <- 40

x <- listings_rdd$centered_overall_score
y <- listings_rdd$num_reviews_l12m
df <- data.frame(x = x, y = y) |> na.omit()

breaks <- seq(min(df$x), max(df$x), length.out = nbins + 1)
df$bin <- cut(df$x, breaks = breaks, include.lowest = TRUE)
bin_centers <- (breaks[-length(breaks)] + breaks[-1]) / 2

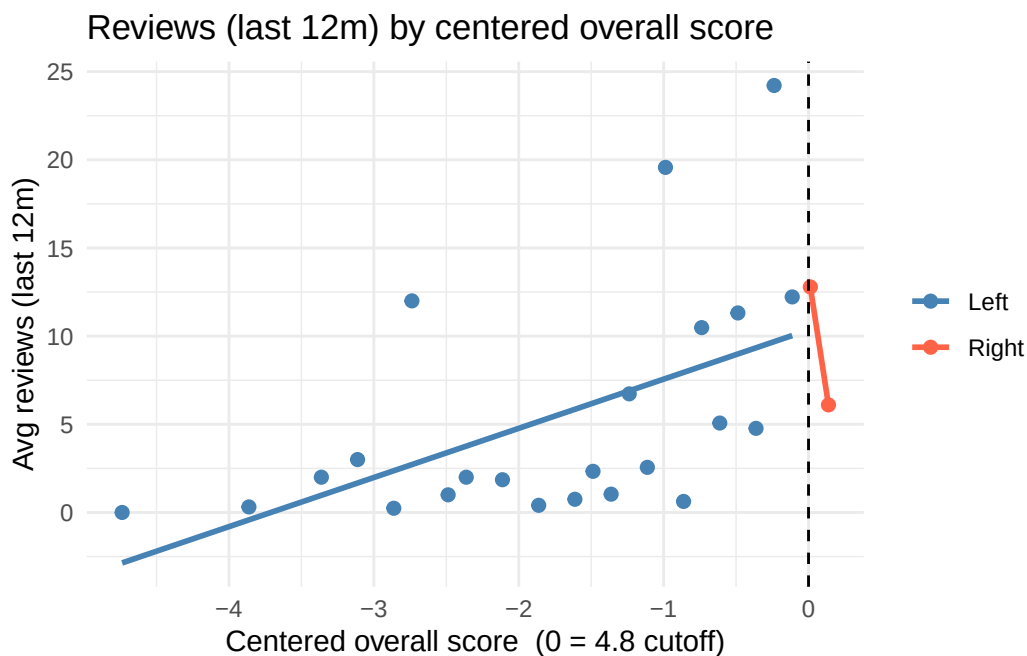
bin_means <- tapply(df$y, df$bin, mean, na.rm = TRUE)
plot_df <- data.frame(
  x = bin_centers,
  mean = as.numeric(bin_means),
  side = ifelse(bin_centers < cutoff, "Left", "Right")
) |> na.omit()

ggplot(plot_df, aes(x = x, y = mean, colour = side)) +
  geom_point(size = 2) +
  geom_smooth(
```

```

    data  = subset(plot_df, side == "Left"),
    method = "lm", formula = y ~ x, se = FALSE
  ) +
  geom_smooth(
    data  = subset(plot_df, side == "Right"),
    method = "lm", formula = y ~ x, se = FALSE
  ) +
  geom_vline(xintercept = cutoff, linetype = "dashed", colour = "black") +
  scale_colour_manual(values = c("Left" = "steelblue", "Right" = "tomato")) +
  labs(
    title = "Reviews (last 12m) by centered overall score",
    x      = "Centered overall score (0 = 4.8 cutoff)",
    y      = "Avg reviews (last 12m)",
    colour = NULL
  ) +
  theme_minimal()

```



```

ggplot(listings_rdd, aes(x = avg_host_review_score)) +
  geom_histogram(binwidth = 0.05, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 4.8, linetype = "dashed", colour = "red") +
  labs(
    title = "Distribution of host average review score",

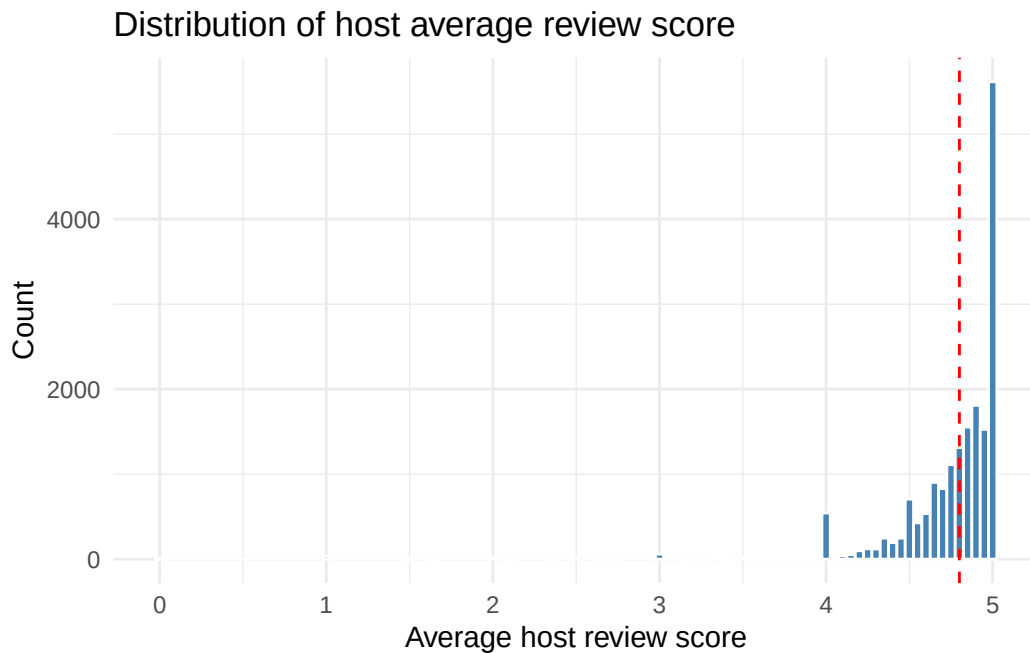
```

```

  x      = "Average host review score",
  y      = "Count"
) +
theme_minimal()

```

Warning: Removed 5997 rows containing non-finite outside the scale range (`stat\_bin()`).



Fuzzy RDD Model

The fuzzy design uses `has_superhost` as the treatment indicator. Crossing the 4.8 threshold acts as an instrument — not everyone above the cutoff gets superhost status (and a few below do), so we estimate the LATE via an IV-like 2SLS approach.

```

rdd_result <- rdrobust(
  y      = listings_rdd$num_reviews_l12m,
  x      = listings_rdd$centered_overall_score,
  fuzzy  = listings_rdd$has_superhost,
  masspoints = "adjust",
  all    = TRUE
)

```

Warning in rdrobust(y = listings_rdd\$num_reviews_l12m, x = listings_rdd\$centered_overall_score, : Mass points detected in the running variable.

```
summary(rdd_result)
```

Fuzzy RD estimates using local polynomial regression.

Number of Obs.	18292	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
Number of Obs.	6948	11344
Eff. Number of Obs.	1269	1771
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	0.059	0.059
BW bias (b)	0.105	0.105
rho (h/b)	0.559	0.559
Unique Obs.	607	194

First-stage estimates.

Method	Coef.	Std. Err.	z	P> z	[95% C.I.]
Conventional	-0.148	0.037	-4.042	0.000	[-0.219 , -0.076]
Bias-Corrected	-0.170	0.037	-4.649	0.000	[-0.242 , -0.098]
Robust	-0.170	0.043	-3.940	0.000	[-0.255 , -0.085]

Treatment effect estimates.

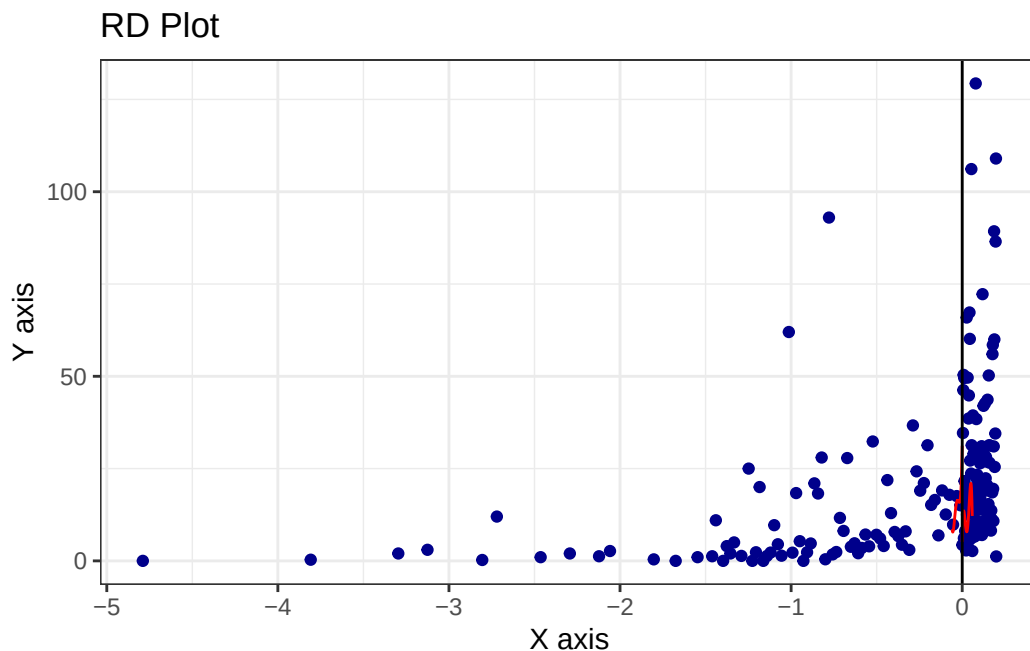
Method	Coef.	Std. Err.	z	P> z	[95% C.I.]
Conventional	95.222	32.824	2.901	0.004	[30.888 , 159.557]
Bias-Corrected	96.933	32.824	2.953	0.003	[32.599 , 161.268]
Robust	96.933	39.572	2.450	0.014	[19.373 , 174.494]

RD Plot

Visualise the fit using the MSE-optimal bandwidth selected by the model.

```
rdplot(  
  y      = listings_rdd$num_reviews_l12m,  
  x      = listings_rdd$centered_overall_score,  
  h      = rdd_result$bws["h", ],  
  masspoints = "adjust"  
)
```

```
[1] "Mass points detected in the running variable."
```



The robust results show that we are 95% confident that for the hosts who, when crossing the 4.8 overall rating threshold, would have received superhost status, the effect of becoming a superhost would have increased their number of reviews somewhere between 20 and 165. Since number of reviews is a proxy for bookings, this effect could translate to even more actual bookings.

Overall though, we can't conclude anything. Our falsification tests suggest that the assumptions needed for our Fuzzy RDD did not hold. Therefore, our estimate has a chance of being biased!

Specification Robustness

Does the LATE estimate change meaningfully when we use a different functional form? We compare three specifications: linear local polynomial (the default), quadratic local polynomial, and a Gaussian Process model (Cho, Kim & Hazlett, 2024; arXiv:2407.10442). Consistency across specifications strengthens the causal claim.

```
# --- Quadratic local polynomial (p=2) ---
rdd_quad <- rdrobust(
  y      = listings_rdd$num_reviews_l12m,
  x      = listings_rdd$centered_overall_score,
  fuzzy  = listings_rdd$has_superhost,
  p      = 2,
  masspoints = "adjust",
  all    = TRUE
)
```

Warning in rdrobust(y = listings_rdd\$num_reviews_l12m, x = listings_rdd\$centered_overall_score, : Mass points detected in the running variable.

```
# --- Gaussian Process (fuzzy via reduced form / first stage) ---
h_opt <- rdd_result$bws["h", 1]
gp_mask <- !is.na(listings_rdd$centered_overall_score) &
  !is.na(listings_rdd$num_reviews_l12m) &
  abs(listings_rdd$centered_overall_score) <= h_opt
gp_df <- listings_rdd[gp_mask, ]

# Reduced form: Y ~ X
gp_rf <- gp_rdd(X = gp_df$centered_overall_score,
  Y = gp_df$num_reviews_l12m,
  cut = 0)

# First stage: D ~ X
gp_fs <- gp_rdd(X = gp_df$centered_overall_score,
  Y = gp_df$has_superhost,
  cut = 0)

# Fuzzy LATE = reduced form gap / first stage gap
gp_late <- gp_rf$tau / gp_fs$tau
# Delta method SE for the ratio
```

```

gp_se      <- abs(gp_late) * sqrt((gp_rf$se / gp_rf$tau)^2 +
                                   (gp_fs$se / gp_fs$tau)^2)

specs_df <- data.frame(
  spec      = c("Linear (p=1)", "Quadratic (p=2)", "Gaussian Process"),
  estimate  = c(rdd_result$coef[1], rdd_quad$coef[1], gp_late),
  ci_lo     = c(rdd_result$ci[3, 1], rdd_quad$ci[3, 1], gp_late - 1.96 * gp_se),
  ci_hi     = c(rdd_result$ci[3, 2], rdd_quad$ci[3, 2], gp_late + 1.96 * gp_se)
)
specs_df$spec <- factor(specs_df$spec, levels = rev(specs_df$spec))

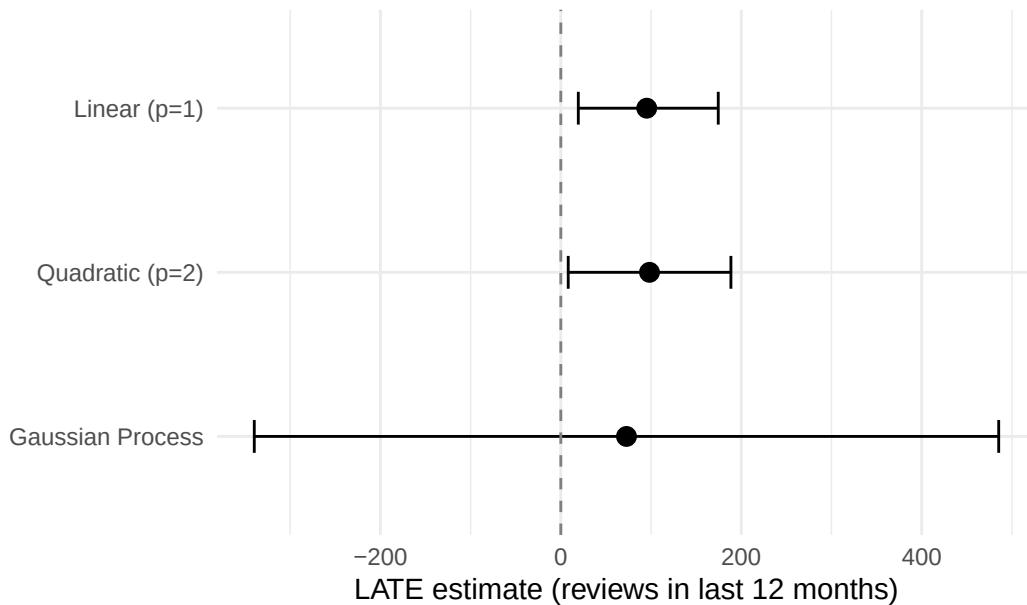
ggplot(specs_df, aes(x = estimate, y = spec)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = ci_lo, xmax = ci_hi), height = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "gray50") +
  labs(
    title = "Specification Robustness: Fuzzy RDD LATE Estimates",
    x      = "LATE estimate (reviews in last 12 months)",
    y      = NULL
  ) +
  theme_minimal()

```

Warning: `geom\_errorbarh()` was deprecated in ggplot2 4.0.0.
 i Please use the `orientation` argument of `geom\_errorbar()` instead.

`height` was translated to `width`.

Specification Robustness: Fuzzy RDD LATE Estimates



Covariate Balance Checks

If the RDD design is valid, pre-treatment covariates should not jump at the cutoff. For each covariate we compute bootstrapped t-statistics under three specifications: full-sample OLS, discontinuity-sample OLS (within the MSE-optimal bandwidth), and local linear regression via `rdrobust`. All t-statistics use bootstrap standard errors (500 resamples).

```
covariates <- c("qualified_by_response_rate", "prop_instant_bookable",  
               "host_tenure_days")  
h_opt <- rdd_result$bws["h", 1]  
  
set.seed(42)  
balance_results <- do.call(rbind, lapply(covariates, function(cov) {  
  df_bal <- listings_rdd[!is.na(listings_rdd[[cov]]) &  
    !is.na(listings_rdd$centered_overall_score), ]  
  
  fml <- reformulate(c("centered_overall_score", "has_superhost",  
                      "centered_overall_score:has_superhost"),  
                    response = cov)  
  
  # Original full-sample coefficient  
  orig_full <- coef(lm(fml, data = df_bal))["has_superhost"]
```

```

# Bootstrap SE - full sample
boot_full <- replicate(500, {
  idx <- sample(nrow(df_bal), replace = TRUE)
  coef(lm(fml, data = df_bal[idx, ]))["has_superhost"]
})
t_full <- orig_full / sd(boot_full)

# Original discontinuity-sample coefficient
df_disc <- df_bal[abs(df_bal$centered_overall_score) <= h_opt, ]
orig_disc <- coef(lm(fml, data = df_disc))["has_superhost"]

# Bootstrap SE - discontinuity sample
boot_disc <- replicate(500, {
  idx <- sample(nrow(df_disc), replace = TRUE)
  coef(lm(fml, data = df_disc[idx, ]))["has_superhost"]
})
t_disc <- orig_disc / sd(boot_disc)

# Local linear (rdrobust, sharp - covariate as outcome)
rdd_cov <- rdrobust(y = df_bal[[cov]],
  x = df_bal$centered_overall_score,
  masspoints = "adjust")
t_local <- rdd_cov$coef[1] / rdd_cov$se[3] # conventional coef / robust SE

data.frame(covariate = cov,
  t_full = abs(t_full),
  t_disc = abs(t_disc),
  t_local = abs(t_local))
}))

```

```

Warning in rdrobust(y = df_bal[[cov]], x = df_bal$centered_overall_score, :
Mass points detected in the running variable.
Warning in rdrobust(y = df_bal[[cov]], x = df_bal$centered_overall_score, :
Mass points detected in the running variable.
Warning in rdrobust(y = df_bal[[cov]], x = df_bal$centered_overall_score, :
Mass points detected in the running variable.

```

```
balance_results
```

	covariate	t_full	t_disc	t_local
has_superhost	qualified_by_response_rate	84.908148	42.138174	4.081870743

has_superhost1	prop_instant_bookable	5.711424	5.974335	0.006577506
has_superhost2	host_tenure_days	13.716561	9.012899	0.892799373

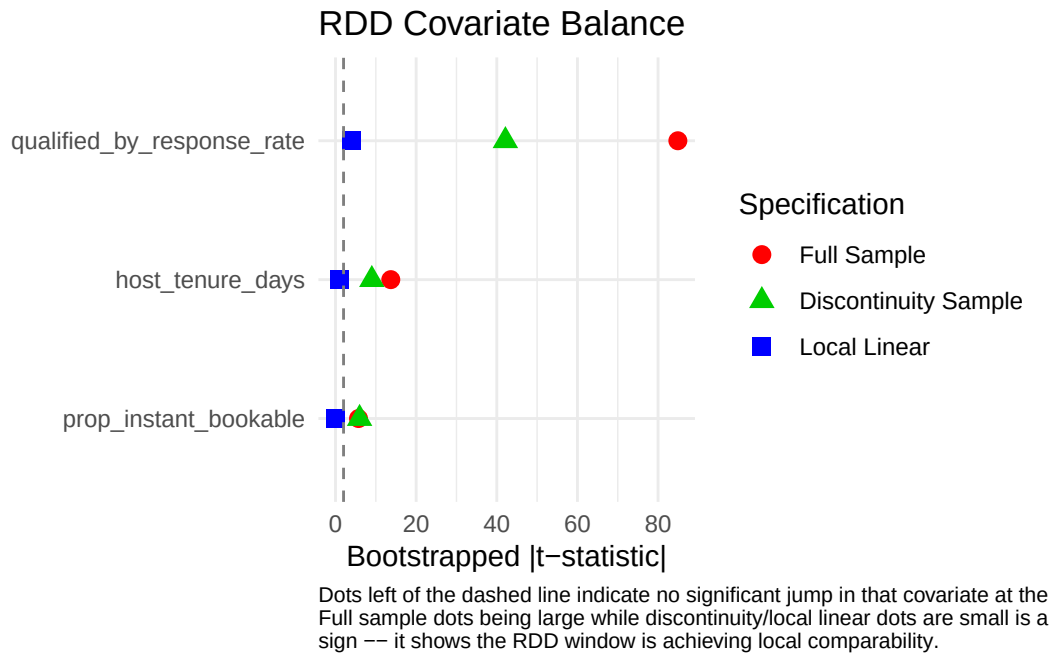
```
balance_long <- pivot_longer(
  balance_results,
  cols      = c(t_full, t_disc, t_local),
  names_to  = "spec",
  values_to = "t_stat"
)

balance_long$spec <- factor(
  balance_long$spec,
  levels = c("t_full", "t_disc", "t_local"),
  labels = c("Full Sample", "Discontinuity Sample", "Local Linear")
)

# Order covariates by full-sample t ascending (bottom to top)
balance_results <- balance_results[order(balance_results$t_full), ]
balance_long$covariate <- factor(balance_long$covariate,
                                levels = balance_results$covariate)

ggplot(balance_long, aes(x = t_stat, y = covariate,
                        colour = spec, shape = spec)) +
  geom_point(size = 3) +
  geom_vline(xintercept = 2, linetype = "dashed", colour = "gray50") +
  scale_colour_manual(values = c("Full Sample"      = "red",
                                "Discontinuity Sample" = "green3",
                                "Local Linear"      = "blue")) +
  scale_shape_manual(values = c("Full Sample"      = 16,
                                "Discontinuity Sample" = 17,
                                "Local Linear"      = 15)) +
  labs(
    title   = "RDD Covariate Balance",
    x       = "Bootstrapped |t-statistic|",
    y       = NULL,
    colour  = "Specification",
    shape   = "Specification",
    caption = str_wrap(
      "Dots left of the dashed line indicate no significant jump in that covariate at the cut"
    )
  ) +
  theme_minimal() +
```

```
theme(plot.caption = element_text(hjust = 0, size = 8))
```



Placebo Cutoffs

If the effect we estimated is truly causal, moving the cutoff to a value where no policy threshold exists should produce a null effect. We test placebo cutoffs at the 25th percentile (4.68) and median (4.87) of the running variable; the true cutoff (4.80) is included for reference. Note: the 75th percentile equals 5.0 (the score ceiling), making RDD infeasible there.

```
cutoffs <- data.frame(
  label = c("4.68 (p25, placebo)", "4.80 (true cutoff)", "4.87 (median, placebo)"),
  cutoff = c(4.68, 4.80, 4.87),
  is_true = c(FALSE, TRUE, FALSE)
)

placebo_results <- do.call(rbind, lapply(seq_len(nrow(cutoffs)), function(i) {
  x_recentered <- listings_rdd$avg_host_review_score - cutoffs$cutoff[i]

  rdd_p <- rdrobust(
    y = listings_rdd$num_reviews_l12m,
    x = x_recentered,
```

```

    fuzzy      = listings_rdd$has_superhost,
    masspoints = "adjust",
    all        = TRUE
  )

  data.frame(
    label      = cutoffs$label[i],
    estimate   = rdd_p$coef[1],
    ci_lo      = rdd_p$ci[3, 1],
    ci_hi      = rdd_p$ci[3, 2],
    is_true    = cutoffs$is_true[i]
  )
}))

```

Warning in rdrobust(y = listings_rdd\$num_reviews_l12m, x = x_recentered, : Mass points detected in the running variable.
Warning in rdrobust(y = listings_rdd\$num_reviews_l12m, x = x_recentered, : Mass points detected in the running variable.
Warning in rdrobust(y = listings_rdd\$num_reviews_l12m, x = x_recentered, : Mass points detected in the running variable.

```
placebo_results
```

	label	estimate	ci_lo	ci_hi	is_true
1	4.68 (p25, placebo)	329.3681	-198.9775	854.1155	FALSE
2	4.80 (true cutoff)	95.2222	19.3730	174.4936	TRUE
3	4.87 (median, placebo)	-292.0478	-1926.2609	1126.1188	FALSE

```

placebo_results$label <- factor(placebo_results$label, levels = placebo_results$label)

ggplot(placebo_results, aes(x = estimate, y = label,
                           colour = ifelse(is_true, "True Cutoff", "Placebo"))) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = ci_lo, xmax = ci_hi), height = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "gray50") +
  scale_colour_manual(values = c("True Cutoff" = "black", "Placebo" = "gray50")) +
  labs(
    title      = "Placebo Cutoff Test",
    subtitle   = "Placebo cutoffs should show no significant effect (CI crosses 0)",
    x          = "LATE estimate (reviews in last 12 months)",
    y          = NULL,

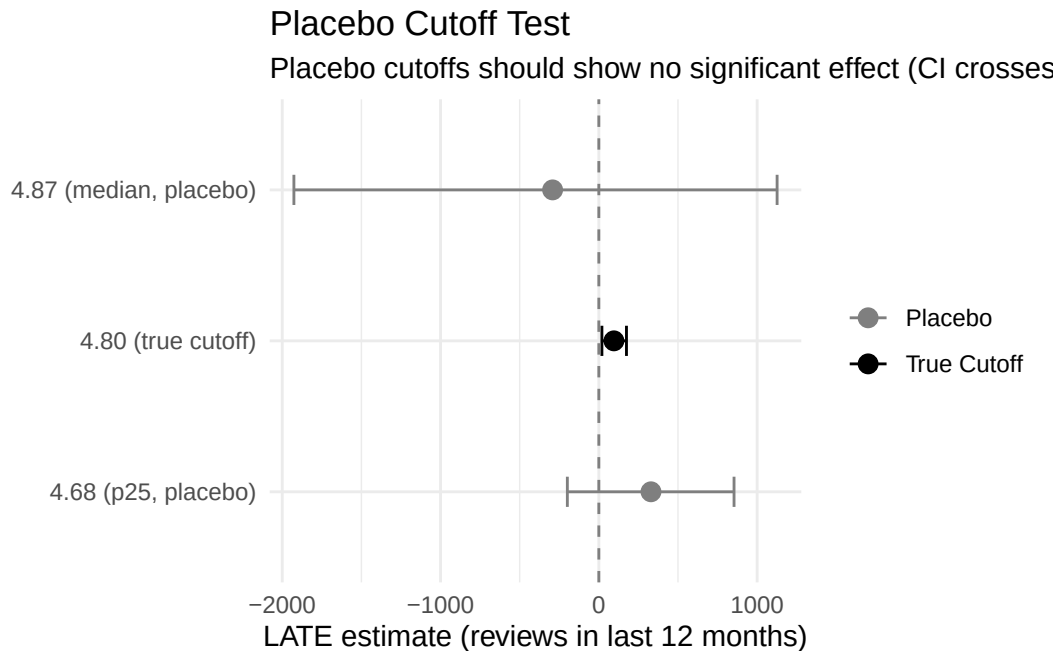
```

```

    colour = NULL
  ) +
  theme_minimal()

```

`height` was translated to `width`.



Falsification test for sorting

Check for sorting/manipulation at the cutoff. Excess mass just to the right of 0 would indicate hosts gaming their scores to cross the 4.8 threshold.

```

density_test <- rddensity(X = listings_rdd$centered_overall_score, c = 0, massPoints = TRUE)

```

Warning in rddensity(X = listings_rdd\$centered_overall_score, c = 0, massPoints = TRUE): 599

```

summary(density_test)

```

Manipulation testing using local polynomial density estimation.

Number of obs =	18292	
Model =	unrestricted	
Kernel =	triangular	
BW method =	estimated	
VCE method =	jackknife	
c = 0	Left of c	Right of c
Number of obs	6948	11344
Eff. Number of obs	469	817
Order est. (p)	2	2
Order bias (q)	3	3
BW est. (h)	0.023	0.023
Method	T	P > T
Robust	10.4191	0

Warning in summary.CJMrddensity(density_test): There are repeated observations.
Point estimates and standard errors have been adjusted. Use option
massPoints=FALSE to suppress this feature.

P-values of binomial tests ($H_0: p=0.5$).

Window Length / 2	<c	>=c	P> T
0.004	23	344	0.0000
0.006	32	363	0.0000
0.008	43	371	0.0000
0.010	204	539	0.0000
0.012	219	550	0.0000
0.015	239	573	0.0000
0.017	254	590	0.0000
0.019	266	601	0.0000
0.021	454	807	0.0000
0.023	469	817	0.0000

```
rdplotdensity(
  rdd      = density_test,
  X        = listings_rdd$centered_overall_score,
  xlabel   = "Centered overall score"
)
```

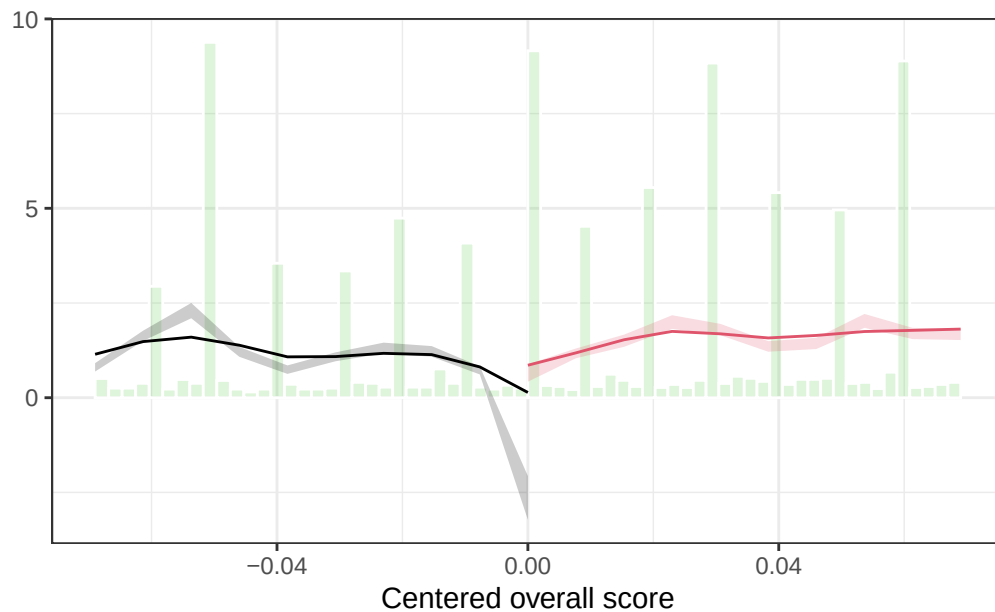
Warning in rdplotdensity(rdd = density_test, X = listings_rdd\$centered_overall_score, : 5997

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

i Please use `linewidth` instead.

i The deprecated feature was likely used in the lpdensity package.

Please report the issue to the authors.



\$Est1

Call: lpdensity

Sample size	6948
Polynomial order for point estimation (p=)	2
Order of derivative estimated (v=)	1
Polynomial order for confidence interval (q=)	3
Kernel function	triangular
Scaling factor	0.379858947023126
Bandwidth method	user provided

Use summary(...) to show estimates.

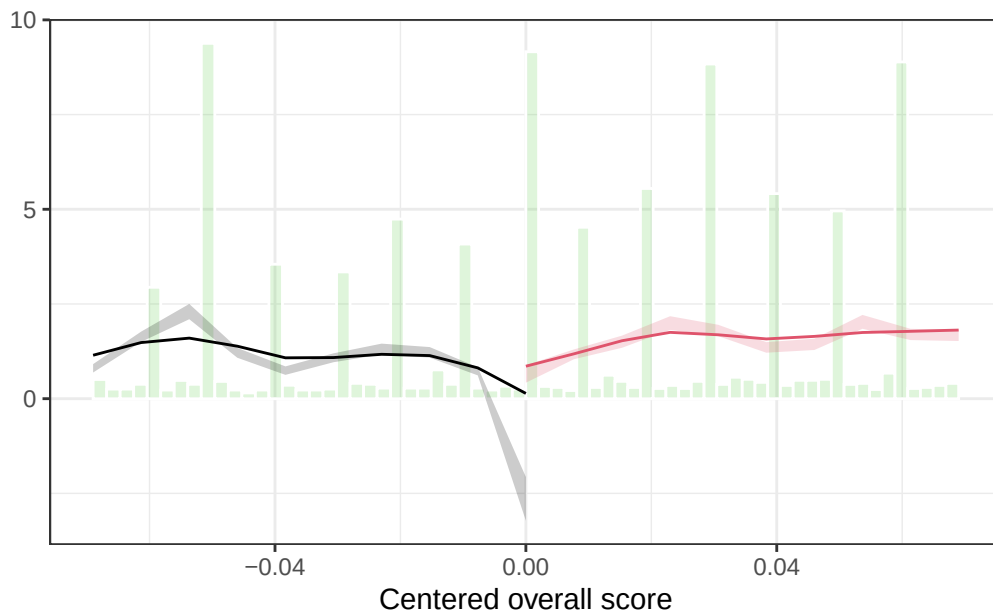
\$Estr

Call: lpdensity

Sample size		11344
Polynomial order for point estimation	(p=)	2
Order of derivative estimated	(v=)	1
Polynomial order for confidence interval	(q=)	3
Kernel function		triangular
Scaling factor		0.620195724673337
Bandwidth method		user provided

Use `summary(...)` to show estimates.

`$Estplot`



H_0 : The density of centered scores is continuous at the cutoff (no evidence of sorting)

H_1 : The density of centered scores is discontinuous at the cutoff (evidence of sorting)

For all the window lengths, the results suggest that the density of overall ratings is discontinuous around the cutoff.

The trustworthiness of our results is actually ambiguous. According to this test, there could have been sorting. Maybe hosts found a way to inflate their rating with fake reviews since the 4.8 cutoff is one of the criteria to become a superhost that is public.

At the same time it's possible that the differences in density on either side of the cutoff are just because the cutoff is at 4.8 and the highest rating possible is a 5.

Because the data could:

- have hosts that are mostly above the cutoff - be subject to sorting

The results of the test are unreliable. This heavily affects the validity of our design.

Fuzzy RDD Intuition

Why fuzzy instead of sharp?

In a sharp design: outcome $\sim$ running $\times$ treatment gives the LATE directly.

The problem with naïve estimation: - Y_1 appears smaller than it should be because some units above the threshold are not treated (didn't get superhost despite score ≥ 4.8) - Y_0 appears larger than it should be because some units below the threshold are treated (got superhost despite score < 4.8) - $\rightarrow$ The effect is underestimated

The fix: Use the threshold indicator Z (above/below 4.8) as an instrumental variable and run a 2SLS-style regression:

- 1st stage (first stage): $D \sim Z \times R$
- Reduced form: $Y \sim Z \times R$
- LATE = reduced form / first stage